Designing Lattice-Based Sequential Aggregate Signature Scheme for Securing Consumer Electronics-Centric IoMT

Sunil Prajapat, Deepika Gautam, Pankaj Kumar, Ashok Kumar Das, *Senior Member, IEEE*, and M.Shamim Hossain, *Senior Member, IEEE*

***Abstract*—The Consumer-Centric Internet of Medical Things (CCIoMT) is an assortment of clinical hardware and Internet of Things (IoT) that garners real-time data through remote monitoring and reduces doctor-consumer or patient interaction, enhancing the quality of remedies for the healthcare system. However, the facile availability and accessibility of medical data on the Web have become a significant cause for information security and trustworthiness disputes in CCIoMT. The computerized algorithm employs a cryptography strategy to accomplish security services like protection, trustworthiness, and non-repudiation. Recently, researchers have developed a variety of advanced signature algorithms for CCIoMT. However, the existing advanced schemes depend on traditional difficult numerical problems like discrete logarithm problems and computational Diffie-Hallman problems, which are unable to safeguard them from rising quantum assaults and some have high computational expanse issues. Lattice-based cryptographic algorithms are one of the solutions to quantum assaults. Therefore, this article portrays a certificateless sequential aggregate signature scheme with the assistance of the smallest integer solution (SIS) and shortest vector problem (SVP) of the NTRU lattices for the CCIoMT network. The proposed scheme aims to provide efficient security solutions for resource-constrained consumer electronics within CCIoMT environments. The portrayed signature scheme's security and operability are shown through security analysis and the performance assessment with some pre-existing mechanisms. The results indicate that the proposed algorithm is suitable for CCIoMT.**

***Index Terms*—Consumer electronic, Internet of Medical Things (IoMT), sequential aggregate signature, communication security.**

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Sunil Prajapat is with the Department of Computer Engineering, AI Security Research Center, Gachon University, Seongnam 13120, South Korea (e-mail: sunilprajapat645@gmail.com).

Deepika Gautam and Pankaj Kumar are with the Srinivasa Ramanujan Department of Mathematics, Central University of Himachal Pradesh, Dharamshala, 176215, India (e-mail: gautamdeepika1999@gmail.com; pkumar240183@gmail.com).

Ashok Kumar Das is with the Center for Security, Theory and Algorithmic Research, International Institute of Information Technology, Hyderabad 500032, India, and also with the Department of Computer Science and Engineering, College of Informatics, Korea University, Seoul 02841, South Korea (e-mail: iitkgp.akdas@gmail.com).

M. Shamim Hossain is with the Research Chair of Pervasive and Mobile Computing, Department of Software Engineering, College of Computer and Information Sciences, King Saud University, Riyadh 13273, Saudi Arabia (e-mail: mshossain@ksu.edu.sa).

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I. Introduction

THE CIRCUMSTANCES surrounding the Corona virus have led to the refurbishment of the healthcare domain into an e-healthcare system. Concurrently, the IoT market in medical care has also garnered attention. The Consumer-Centric Internet of Medical Things (CCIoMT) is a collaboration between electronic health systems and the Internet of Things in healthcare. The CCIoMT network offers plenteous superiorities over conventional healthcare systems, for instance, remote and continuous monitoring and cost-effective and enhanced remedies. The scientific potential of CCIoMT improves individuals lives with its multiple benefits. CCIoMT's patient-centered care approach not only curtails doctor-consumers interaction, but is also more approachable and less time-consuming. Additionally, the CCIoMT oversees the insightful administration of patients, clinical supplies, and faculty. The structural design of the CCIoMT network is comprised of sensors or medical devices, a cloud server, a consumer or patient and medical faculty. The number of discrete sensors or medical devices worn or embedded into the patient's body that accumulate real-time data for diagnosis and treatment [1], [2], [3]. The accumulated data is stowed, analyzed, and processed on the medical cloud server and is enabled to be securely accessible to everyone in the network. Consequently, medical professionals can suggest an early and swift diagnosis with improved prescriptions by analyzing the consumer's real-time medical data on a cloud server.

Unfortunately, the facile availability and accessibility of medical data on the Web has become a significant cause of unauthorized access, information security, and data breach disputes in the CCIoMT. Any tempering or deletion of a consumer's severity and sensitive information by a pernicious individual can have life-threatening consequences. For instance, consider a scenario where a medical care administration is continuously treating a diabetic patient, and a malicious individual alters the information to include a cancer patient. The medication given to the diabetic patient may result in the patient's death. As a result, contextual privacy and data security have emerged as a primary concern in the network [4], [5], [6]. For better functionality and to boost credibility in the CCIoMT network, non-repudiation, availability, data privacy, authentication, and integrity are some required security services. Therefore, computerized algorithms

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relying on cryptography, such as digital signatures, are the best solution to fulfill the security requirements of networks.

A digital signature is a cryptographic tool used to ratify the legitimacy of messages through mathematical schemes. Plenteous of signature protocols are in the literature for the CCIoMT network. However, the traditional individual corroboration of signatures for large scale environments like CCIoMT demands high computational expenditure. This one-by-one approach prompts a significant computational burden, hindering efficiency and scalability. As a result, it becomes a bottleneck for network deployment. Therefore, the resource-restraint environments of IoT applications such as CCIoMT employ the aggregate signature technique of cryptography to reduce computational expenditure and improve security. An aggregate signature is capable of realizing non-repudiation on 'n' different signatures concurrently by compressing them into a single signature [7], [8], [9]. On the other side, a sequential aggregate signature is a kind of aggregate signature that aggregates the signature in sequence and is similar to the layer in onion. However, distingushing from the aggregate signature, 'n' signatures are not generated for the sequential aggregate signature. On the contrary, a potential signer incorporates a signature on the message of his selection into one sequential aggregate to transmute it to become a new sequential aggregate [10]. Sequential aggregate signatures offer better advantages over aggregated signatures, such as granular control, enhanced scalability, efficiency, and security for the CCIoMT network, while reducing computational and communication expenditures. Subsequently, a sequential aggregate signature improves authentication security by aggregating each new signature with the previous one to create a verifiable chain of signatures. As a result, the inclusion of sequential aggregate signatures in CCIoMT is a significant advancement that improves the computational efficiency and security of signature verification processes.

A. Motivation and Contribution

Numerous advanced signature protocols have emerged in response to the aforementioned disputes. Lamentably, these protocols encounter numerous constrictions, such as key escrow issues and certificate supervision and some are computationally wasteful or are vulnerable to various security attacks. Likewise, security of these protocols relies on conventional complex mathematical assumptions, i.e., the Diffie-Hallman problem, the discrete logarithm problem, and the greatest integer factorization. The quantum computer can easily unravel the associated complexity of the problem's solution. Therefore, these protocols are unable to offer the required security against quantum assault. Additionally, given the limited resources, guaranteeing security, authentication, and privacy in consumer electronics continues to be a difficult task. Many times with great processing cost, traditional encryption methods are not fit for lightweight consumer devices. The assistance of the certificateless cryptosystem and lattice cryptography, along with the benefits of sequential aggregate signature are able to resolve the issues encountered with CCIoMT simultaneously. This has motivated us to develop a certificateless lattice-based signature scheme employing the sequential aggregate signature to obtain proficient security in the CCIoMT network. The main research contributions are listed below:

- We proposed a lightweight, certificateless sequential aggregate signature scheme specifically designed for an IoMT environment. This protocol utilizes the benefits of a certificateless cryptosystem. As a result, the protocol does not encounter key escrow or certificate management issues with ID-based and public-key cryptosystems.
- We added a lattice-based cryptosystem to the CCIoMT network that provides scalability, confidentiality, privacy, and resistance to quantum threats in order to make the proposed protocol more efficient and safe. We achieve this security by using the SIS lattice problem and one-way hash functions for communication. Mathematical analysis and the Real or Random Oracle Model (ROR) demonstrate the protocol's competency against various attacks, thereby validating its security and effectiveness.
- The proposed protocol employs a sequential aggregate signature, which reduces the computational expanses of the resource constraint environment of CLSAS while also improving the protocol's security. Furthermore, we also provided a performance evaluation of the presented CLSAS (certificateless sequential aggregate signature) scheme and demonstrated that the designed protocol is more computationally efficient and secure than the other existing protocols.

B. Structure of Article

The subsequent structure of this article is organized as follows: Section II pertains to the literature review or related works. Section III comprises preliminaries that encompass lattices and their associated complex problems. Section IV addresses the framework concept, syntax, and requirements for security. Section V presents the proposed strategy for CLSAS. Section VI presents the security analysis of the proposed scheme. Section VII encompasses the evaluation of the proposed work's efficiency and a comparison analysis of its performance against pertinent rival plans. Finally, in Section VIII, we complete the study.

II. Related Work

This section highlights the contributions and limitations of some existing state-of the-art. This section is further divided into subsections: a) aggregate and sequential signature; b) lattice-based signature.

A. Aggregate and Sequential Signature

The digital signature system was first broached in 1976 and has been used for more than 40 years. Exploiting the attributes of a public-key cryptosystem, abundant digital signatures prevail in the literature. Unfortunately, managing certificates associated with a consumer's public key is a chief dispute in this cryptosystem. Shamir [11] brought up an identity-based cryptosystem in 1984 that eradicates the requisite certificates. Though the cryptosystem has given rise to the key escrow difficulty, Later, to exterminate this, a certificateless cryptosystem was projected by Al-Riyami and Paterson [12]

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in 2003. This cryptosystem concurrently holds the attributes of the public key and identity-based cryptosystems.

Boneh et al. [13] came up with the idea of aggregate signature in 2003. Aggregate signatures are convenient for reducing computational expenses as well as enhancing the competence of the resource-restrained environment. The advent of an aggregate signature has directed the digital signature to a new avenue. Subsequently, researchers devise numerous aggregate signature schemes, either with or without pairing operations. The work done employs bilinear maps and is devised with elliptic curve cryptography.

Lysyanskaya et al. [14] projected a new type of aggregate signature termed sequential aggregate signature in 2004. They used the trapdoor permutation to design the signature scheme, but their scheme has two shortcomings. In their work, the individual signer should verify an aggregate signature to insert their signature into it, and trapdoor permutation must be certified [15]. With an interactive assumption, an identity-based sequential aggregate signature protocol is broached under the random oracle model by Boldyreva et al. [16]. Unfortunately, their work is found vulnerable by Hwang et al. [17]. Lu et al. [18] also presented a sequential aggregate signature. In their work, it is not essential for verification to know the sequence in which messages were signed for the aggregate signature. Utilizing the short public key, the sequential aggregate signature scheme was proposed by Lee et al. [19]. They used prime-order bilinear groups for the short public keys and demonstrated the security under a standard model with static assumptions.

B. Lattice-Based Signature

Implementing Shor's algorithm on the quantum computer and the vulnerability of conventional protocols against it have escalated the requirement for quantum resilient algorithms. The exploitation of lattice-based cryptography for these quantum-resilient algorithms is an answer to this problem. The Ajtai put forward the first lattice-based algorithm [20] in 1996. He utilized the shortest integer solution (SIS) problem in his algorithm. An NTRU lattice-based certificateless signature protocol was projected by Xie et al. [21] with the SIS problem. Hung et al. [22] also broached the lattice-based certificateless signature scheme with a revocation method to revoke the malicious. Later, evidence of their vulnerabilities against the type 1 and type 2 adversaries was presented by the KYUNG-AH SHIM [23].

With the employment of the lattice-based signature, an aggregate signature scheme was constructed by Doroz et al. [24]. Later, Boneh and Kim [25] broached two novel aggregate signature schemes on the premises of lattices. When using these protocols, such as one-time Aggregate Sequential Signatures (ASS) and lattice-based ASS, the size of the aggregate signature is at most logarithmic in the number of signatures being aggregated. Finally, Qian et al. [26] presented a lattice-based data aggregation scheme employing aggregate signature and the homomorphic cryptosystem. They stated that their scheme could withstand a quantumm attack because of the utilized lattice hard problem and homomorphic cryptography. Furthermore, they constructed their aggregate signature using a new Batch RSA algorithm. Brogle et al. [15] broached a sequential aggregate signature scheme employing the trapdoor permutation. In contrast to other sequential aggregate signature mechanisms, their mechanism does not require the verification of the aggregate by each signer before including their signature and message in the aggregate. They have acquired it through the utilization of the lazy verification technique. A sequential aggregate signature based on lattices is projected by Jia et al. [27], and their security depends on the NTRU lattice problem. Additionally, their scheme exploits the benefits of a certificateless cryptosystem and demonstrates security under the random oracle model. For secure logging, Kim et al. [28] brought up a forward secure sequential aggregate signature protocol utilizing the hardness factoring assumption of lattices. The construction of the scheme under the random oracle model employs the hard lattice problem, enabling security against quantum attacks. They also claimed that the size of their keys and aggregate signatures are constant.

III. Mathematical Preliminaries

This section presents the fundamental mathematical preliminaries for the lattice-based cryptography used in the protocol. These include mathematical definitions of lattices, gaussian over lattices, and basic hardness assumptions.

A. Notations

We carefully present the required definitions and findings in this section. If you are interested in more information, please go to [27]. We are working on the rings $R = \frac{Z[x]}{(x^N+1)}$ and $R_p = \frac{Z[x]}{(x^N+1)}$, where p is a prime number greater than 3, respectively. Additionally, the condition that x^N+1 can split into k_p invariant elements modulo prime p, where $k_p = 3$ or N, is satisfied.The set of the revertible component in R is denoted as $R^\times$. If the vector v is chosen from R^N, then $||v||$ stands for the vector's Euclidean norm. Let u_j represent the jth column in matrix M, and $||M||$ be defined as $\max_j(||u_j||)$.

B. Hardness Assumption

Definition 1 ($R - SIS_{p,3,\beta}$): The SIS problem can be expressed on the NTRU lattice by setting $R = \frac{Z[x]}{(x^N+1)}$, choosing tiny polynomials f, g in accordance with, and setting $h = g/f$. So, $(R - SIS_{p,3,\beta})$ is find to (z_1, z_2) that fulfills $(z_1, z_2) \in \Lambda_{h,p}$ and $||(z_1, z_2)|| \leq \beta$.

Proposition 1: Assume $N = 2^N, \phi = (x^N + 1)$, $\epsilon > 0$ and p, q are positive integers such that $p \geq \beta\sqrt{N} \cdot \Omega(\log N)$ and $q, \log p \leq poly(N)$. If there is a polynomial-time algorithm A solving $R - SIS_{p,q,\beta}$ with non-negligibility probability, a new algorithm B can be constructed to solve τ-Ideal-SVP in polynomial time with $\tau \geq \beta\sqrt{N}.\Omega(\log N)$ by invoking algorithm A.

Let $q = 3$, according to Proposition 1, $R - SIS_{p,3,\beta}$ problem can be reduced to the τ-Ideal-SVP.

IV. System Model

This section presents a detailed framework model, a syntax model and the necessary security requirements of the proposed protocol.

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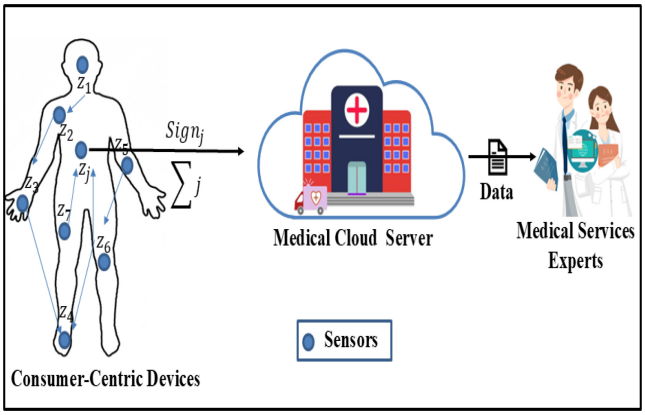

Fig. 1. Consumer Centric IoMT Framework.

A. Framework Model

We utilize the identical system model as described in reference. The primary objective of our system model is to ensure the validity, integrity, and confidentiality of data transfer, thereby eliminating the possibility of erroneous information being transmitted. The system comprises four elements, as seen in Fig. 1 as follows: 3) Medical smart devices (Sensors); 1) Medical service experts; 2) Medical Server (MS); and 4) Aggregator. According to the Care-District provision, each Care-District is responsible for monitoring a certain disease. For example, if we want to monitor heart rate, all the sensors used to measure heart rate will be from the same Care-District.

- *Medical service experts:* Medical service expert with proper authorization utilize medical data collected by sensors that possess certain calculating and communication capabilities. The data is analyzed in order to determine the most suitable treatment for consumers.
- *Medical server (MS):* The MS has robust computational capabilities and sufficient storage capacity, allowing it to effectively process the large amount of data transmitted by sensors. Upon completion of the processing, MS delivers the consumer's information to the healthcare professionals. During the initialization process, MS generates a pair of private and public keys, denoted as MS_{sk} and MS_{pub} respectively. MS then makes the public key, MS_{pub}, available to the public. MS is authorized to produce a partial private key for each sensor and securely send it to the sensors.
- *Medical Smart devices:* Sensors are limited devices with limited resources. Each sensor, identified by ID_j and equipped with its own private-public key pair (SK_j, PK_j), is assumed to belong to a Care-District. Each sensor, identified individually, has the ability to produce a signed message from the physical environment using its own private key SK_j. Subsequently, this signature goes to the next sequential signature aggregator.
- *Aggregator:* An aggregator has limited calculating and computing abilities. The system sequentially collects signatures from a care district, creates an integrated signature, and then sends it to MS. It is assumed that each care district consists of one aggregator and many sensors. Afterwards, it transmits this signature to the subsequent sequential signature aggregator.

The CLSAS construction consists of four entities: sensors on the consumer's body, a MS, a healthcare professional, and an aggregator. MS determines the system parameter, creates a partial private key for sensors based on their ID_js, and securely communicates this partial private key to the sensors through a secure channel. To generate a signature on their mobile devices, the sensors selected their private key. When MS transmits data, it cannot capture the information because it does not have the sensor's private key. In our work, we employ correctness to expedite the process of verifying signatures. This is done for each message in their respective Care-Districts. One of the sensors in the Care-District serves as the aggregator. The system collects all signatures from each Care-District and then uploads the combined signature to the healthcare professional's portable devices over MS. Healthcare experts authenticate the aggregate signatures from which they obtained patient data.

B. Syntax

A CLSAS scheme consists of seven algorithms: Setup, Extract-Partial-key, Set-Secret-Value, Set-Private-key, Set-Public-key, CL-Agg-Sign, and CL-Agg-Verify.

- *Setup* (N)*:* The master private/public key pair (MS_{sk}, MS_{pub}) is produced by MS using the security parameter N as an input. Note that MS protects MS_{sk}'s and holds it private.
- *Extract-Partial-key* (ID_j)*:* MS outputs a partial private key D_j upon input of the identity ID_j of the sensor's SEN_j. Here, $1 \leq j \leq k$.
- *Set-Secret-Value* (ID_j)*:* The sensor's SEN_j outputs a secret value SS_j, given the identity ID_j.
- *Set-Private-Key* (D_j, SS_j)*:* After taking the inputs D_j and SS_j, the sensor's SEN_j outputs SK_j as the private key.
- *Set-Public-Key* (SK_j)*:* On input of a private key SK_j, the sensor's SEN_j outputs a public key PK_j.
- *CL-Agg-Sign* $(Sign_{j-1}, PK_{j-1}, MS_{pub}, \mu_{j-1}, ID_{j-1})$*:* Given the current sensor's SEN_j, the current message μ_j, the previous message set μ_{j-1}, the previous identity set ID_{j-1}, the previous signature set $Sign_{j-1}$, the previous public key set PK_{j-1}, an aggregate signature $Sign_{j-1}$ of the sensor's SEN_{j-1}.
- *CL-Agg-Verify* $(Sign_j, PK_j, MS_{pub}, \mu_j, ID_j)$*:* On input of $Sign_j, PK_j, MS_{pub}, \mu_j, ID_j$, the verifier checks whether $Sign_j$ is a valid certificateless aggregate signature generated by the sensor's SEN_{j-1} on messages μ_j; if so, outputs 1. Otherwise, it outputs 0.

C. Requisites for security

It is necessary to satisfy some of the security attributes by a secure protocol. These are listed below:

- *Correctness:* Our system's generated signature must successfully meet the signature verification criteria for the proposed technique to be valid.
- *Unforgeability:* In the CLSAS security paradigm, the proposed technique is existentially unforgeable against two categories of adversaries.

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1) *Type 1:* An external attacker is represented by the adversary A_1. As a result, every sensor's public key can be replaced by the Type 1 adversary A_1 with a value of his choosing.
2) *Type 2:* A malicious MS is represented by the adversary A_2. As a result, the Type 2 adversary A_2 is aware of the master secret key MS_{sk}.

- *Lightweight:* The suggested approach features low communication and processing costs, making it suitable for application in the real world.

V. The Proposed Scheme on NTRU Lattice: CLSAS

This section addresses the different phases associated with the proposed protocol. The seven stages of our proposed NTRU lattice-based CLSAS system are: setup, partial key extraction, secret value, private key, public key, CL-Agg-Sign, and CL-Agg-Verify.

A. Setup Phase

Suppose $p = \Omega(\beta\sqrt{N})$ and N be a security criterion, η, k, ν be the higher number, $s = \Omega(\sqrt{N\log p}), \sigma = 12s\eta N$, $E_{1,2}, E_{1,1}{:}(0,1)^\star \longrightarrow (0,1)^\ell$, $F_h{:}\{v \in R_p, ||v|| \leq \sigma\sqrt{N}\} \longrightarrow R_p$, $G_{F_h}{:}(0,1)^\ell \longrightarrow \{v \in R_p, ||v|| \leq \sigma\sqrt{\frac{N}{2}}\}$ and $Enc{:}\{0,1\}^\star \longrightarrow \{0,1\}^\star \times \{v \in R_p, ||v|| \leq \sigma\sqrt{\frac{N}{2}}\}$, where k represents the number of signers. $\log\{v \in R_p, ||v|| \leq \sigma\sqrt{\frac{N}{2}}\} > \ell, \ell < N$. MS executes this method after receiving an input of the security parameter N. The subsequent actions are taken:

1) Runs $TrapGen(N, p)$ with the system security parameter N as an input to create a trapdoor B as well as $h \in R_p^\times$ and ensuring $(1+h) \notin R_p^\times$ is stable, where $B = \begin{bmatrix} C(f) & C(g) \\ C(F) & C(G) \end{bmatrix}$, $h = g * f^{-1}$.
2) Establishes $MS_{pub} = h \in R^*$ as MS's public key and $MS_{sk} = B$ as MS's master key.
3) Selects three one-way collision-resistant hash functions:

$$\begin{aligned} H_1&: \quad \{0,1\}^* \longrightarrow \{v \in Z^N\}, \\ H_2&: \quad Z_p^{2N} \times \{0,1\}^* \longrightarrow D_{H_2}{:}\{v \in Z_p^N, 0 \leq ||v||_1 \\ &\quad \leq \eta, \eta << p\}, \\ H_3&: \quad \{0,1\}^* \longrightarrow \{v \in Z^N\}. \end{aligned}$$

4) Keeps the system parameter list $params = (MS_{pub}, H_1, H_2, H_3,)$ public and the master key B secure.

B. Extract-Partial-Key Phase

MS generates the sensor's $SEN_j's$ partial private key and is associated with the sensor's identity ID_j.

1) The MS's master key B and the sensor's identity ID_j are input.
2) The Gaussian sampling algorithm is used to create $D_j = (S_{1,j}, S_{2,j})$ on the NTRU lattice GaussianSampler $(B, s, H_1(ID_j))$. D_j is then used as a partial private key and sent to the appropriate sensor through a secure channel.
3) $SEN_j's$ checks to see if $||(S_{1,j}, S_{2,j})|| \leq s\sqrt{2N}$ and $H_1(ID_j)$ are equal to $S_{1,j} + S_{2,j} * h$ holds. If they hold, set $D_j = (S_{1,j}, S_{2,j})$ as the sensor's partial private key; otherwise, reject.

C. Set-Secret-Value Phase

The secret value is generated by $SEN_j's$ using the identity ID_j.

1) Arbitrary chooses $S'_{1,j}, S'_{j,2} \in D_{Z^N,s}$.
2) Places $SS_j = (S'_{1,j}, S'_{j,2})$ as the sensor's secret value.

D. Set-Private-Key Phase

The entire private key is generated by $SEN_j's$ with the identity ID_j.

1) The partial key D_j and the secret value SS_j are entered.
2) Establishes the $SEN_j's$ whole private key as $SKj = (D_j, SS_j)$.

E. Set-Public-Key Phase

The public key is generated by SEN_j with the identity ID_j.

1) The sensor's complete private key, SKj, is entered.
2) Determine $\delta_j = H_2(ID_j, h, H_1(ID_j)$ & $P_j = S'_{1,j} + S'_{2,j} * h$.
3) Place the sensor's public key as $PK_j = H_1(ID_j) + \delta_j P_j$.

F. CL-Agg-Sign Phase

The sensors added one by one, one by one, all of their signatures on the message $\{\mu_1, \ldots, \mu_j\}$, corresponding to the identities $\{ID_1, \ldots, ID_j\}$ and public keys $\{PK_1, \ldots, PK_j\}$, to make a sequential aggregate signature. The sensor computes $Sign_j$ and generates an output as aggregate signature $\sum_j$ (see Algorithm 1).

G. CL-Agg-Verify Phase

Executing a signature $\sum_j$ of message μ_j on identity ID_j utilizing public key PK_j and state information. If the verification equation is satisfied, accept the signature; otherwise, reject it (see to Algorithm 2).

VI. Security Analysis

In this section, we first provide the correctness proof of our proposed scheme (CLSAS). Next, we provide the detailed security analysis, both formal and informal, of CLSAS.

A. Correctness

The proposed CL-Agg-Sign scheme is only valid if the signature it produces meets the following validity requirements.

$$\begin{aligned} &F_h(sign_{j,1}) - PK_j \cdot c_j = z_{j,1} + z_{j,2} * h - PK_j c_j \\ &= [S_{j,1} + \delta_j S'_{j,1}]c_j + y_{j,1} + [[S_{j,2} + \delta_j S'_{j,2}]c_j + y_{j,2}]h \\ &- [H_1(ID_j) + \delta_j P_j]c_j \\ &= y_{j,1} + y_{j,2} * h. \end{aligned}$$

The valid signature $Sign_j$ obtained using CL-Agg-Sign fulfills the above-mentioned equations.

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Algorithm 1 CL-Agg-Sign

Input: Message μ_j,the $SEN_j's$ fully private key SKj
Output: $\sum_j$

if $j = 0$ **then**
  Set $\sum_0 \leftarrow (0^N, \gamma)$
end if
Determine $\sum_j$ as $(F_h, \vec{\mu_{j-1}}, \vec{\alpha_{j-1}}, sign_{j-1}, c_{j-1}, e_{j-1,1}, e_{j-1,2})$, F_h is the trapdoor function.
if CL-Agg-Verify $(\sum_{j-1}) = \perp$ **then**
  Set $\perp = 1$ or 0
end if
Evaluate $(\alpha_j, \beta_j) \longleftarrow Enc_{F_h}(sign_{j-1})$.
Place $\vec{\alpha_j} = \vec{\alpha_{j-1}}|\alpha_j$,
Evaluate
$e_{j,1} = e_{j-1,1} \oplus E_{11}(F_h, \vec{\mu_{j-1}}, \vec{\alpha_{j-1}}, sign_{j-1})$.
$e_{j,2} = e_{j-1,2} \oplus E_{12}(F_h, \vec{\mu_{j-1}}, \vec{\alpha_{j-1}}, sign_{j-1})$.
Evaluate
$g_{j,1} \leftarrow G_{F_h}(e_{j,1})$, $g_{j,2} \leftarrow G_{F_h}(e_{j,2})$
Evaluate
$y_j = \begin{bmatrix} y_{j,1} \\ y_{j,2} \end{bmatrix} = \begin{bmatrix} g_{j,1} + \beta j \\ g_{j,2} + \beta j \end{bmatrix}$
Evaluate $c_j = H_3([y_{j,1} + y_{j,2} * h], \mu_j)$
Evaluate
$sign_j = \begin{bmatrix} z_{j,1} \\ z_{j,2} \end{bmatrix} = \begin{bmatrix} S_{j,1} + \delta_j S'_{j,1} \\ S_{j,2} + \delta_j S'_{j,2} \end{bmatrix} * C_j + \begin{bmatrix} y_{j,1} \\ y_{j,2} \end{bmatrix}$
$\sum_j \longleftarrow (F_h, \vec{\mu j}, \vec{\alpha j}, sign_j, c_j, e_{j,1}, e_{j,1})$ is come back with possibility $\min(\frac{D_{Z^N,\sigma}(sign_j)}{MD_{Z^N,\sigma,S_j c_j}} sign_j, 1)$.Whenever there is no output, repeat this process.

Algorithm 2 CL-Agg-Verify

Input: Message $\sum_k$
Output $(F_h, \vec{\mu_j})$

Determine $\sum_k$ as $(F_h, \vec{\mu_k}, \vec{\alpha_k}, sign_k, c_k, e_{k,1}, e_{k,2})$
if $j = k \rightarrow 1$ **then**
  if $\log_2(\{v \in R_p, ||v|| \leq \sigma\sqrt{\frac{N}{2}}\}) \leq \ell$, $sign_{j,1}$ or $sign'_{j,1} \notin \{v \in R_p, ||v|| \leq \sigma\sqrt{N}\}$ **then**
    return
  end if
  Evaluate $g_{j,1} \leftarrow G_{F_h}(e_{j,1})$, $g_{j,2} \leftarrow G_{F_h}(e_{j,2})$
  Evaluate $\beta_j = [F_h(sign_{j,1}) - (PK_j) * c_j]$.
  Evaluate $sign_j \leftarrow Dec_{F_h}(\alpha_j, \beta_j)$.
  Evaluate
  $e_{j-1,1} = e_{j,1} \oplus E_{11}(F_h, \vec{\mu_{j-1}}, \vec{\alpha_{j-1}}, sign_{j-1})$.
  $e_{j-1,2} = e_{j,2} \oplus E_{12}(F_h, \vec{\mu_{j-1}}, \vec{\alpha_{j-1}}, sign_{j-1})$.
end if
if $\sum_0 = (0^N, \gamma)$ **then**
  return $(F_h, \vec{\mu_j})$
else
  return $\perp$
end if

B. Formal Analysis

It is shown that the CLSAS scheme can't be broken by strong adversaries in the random oracle model, even if the τ-Ideal-SVP against polynomial-time algorithm is complicated. We provide the proofs of theorems for adversaries 1 and 2 below.

Theorem 1: The new CLSAS method can't be broken by any probabilistic polynomial time (PPT) strong type adversary 1 in the random oracle model if the $(p, 3, (2\sigma + 2\eta s)\sqrt{N})$-SIS on the NTRU lattice $\Lambda_{h,p}$ is impossible to solve.

Proof: Assuming a probabilistic polynomial-time PPT adversary, A_1 exists who has a non-zero probability of defeating the new CLSAS scheme. We can build a challenger C to resolve the SIS problem on the NTRU lattice.

- *Invocation:* The challenger C is called on a random instance of the $(p, 3, \beta) - SIS$ problem on the NTRU lattice $\Lambda_{h,p}$, and it has to return a correct answer.
- *Supplied:* Two polynomials $h, t \in R^*$ and $\Lambda_{h,p}$ of the NTRU lattice.
- *Supplied:* $(a, b) \in \Lambda_{h,p}$
  Assuming that:
  1) For the same input, the following oracles are queried only once.
  2) Prior to any inquiry $Q_k = (F_h, \vec{\mu_k}, \vec{\alpha_{k-1}}, sign_{k-1})$ to $E_{11}\&E_{12}$ and oracles, Oracle query IDj to H must have been made by the sequential forger. Queries $e_{j,1}, e_{j,2}$ to G_{F_h} oracle and queries Q_j to $E_{11}\&E_{12}$ oracle $1 \leq j \leq k-1$.
  3) Sequential forger must have consulted G_{F_h}, E_{11}, E_{12} and H_3 Oracles before making any query $(\sum_j, \mu_j, SK_j)$ to CL-Agg-Sign-Oracle. Additionally, the CL-Agg-Verify $(\sum_j) = 1$.
- *Queries:* A_1 can adaptively ask all the oracles displayed after that.
  1) H_1-oracle-query: challanger C maintains a list $l_{H_1} - list = (ID_j, D_j, S_{j,1} + S_{j,2} * h)$. The identity $ID_j \in \{0, 1\}^*$ is input, and C searches the $l_{H_1} - list$ for it.In the event that ID_j is located in the $l_{H_1} - list$, the challanger C sends A the result $S_{j,1} + S_{j,2} * h$.If not, C selects two polynomials $S_{j,1} + S_{j,2}$ from $D_{Z^N,s}$, saves $(ID_j, D_j = (S_{j,1} + S_{j,2}), S_{j,1} + S_{j,2} * h)$, and then successively returns $S_{j,1} + S_{j,2} * h$ to A_1.
  2) H_2-oracle-query: challanger C maintains a list $l_{H_2} - list = (ID_j, SK_j, \delta j)$. The identity $ID_j \in \{0, 1\}^*$ is input, and C searches the $l_{H_2} - list$ for it.In the event that ID_j is located in the $l_{H_2} - list$, the challanger C sends A_1 the result $\delta j = H_2(ID_j, h, H_1(ID_j))$ saves.
  3) Create-consumer electronic devices-oracle-query: The challanger C maintains a 5-tuple list called $l_c - list$ $(ID_j, D_j, SS_j, SK_j, PK_j)$.Following ID_j input, C executes the following.In the event that ID_j is located in the $l_c - list$, C returns PK_j. Otherwise, the H_1-Oracle-query for D_j and H_2-Oracle-query for $PK_j$is generated by the challanger C. The challanger C will then execute the algorithms Set-Secret-Value, Set-Private-Key, and Set-Public-Key to produce the values $SS_j = (S'_{j,1}, S'_{j,2})$, and $SK_j = (D_j, SS_j)$ and $PK_j = H_1(ID_j) + \delta_j P_j$, where $P_j = S'_{1,j} + S'_{2,j} * h$ respectively. The $l_c - list$ will then be used to hold

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the tuples $(ID_j, D_j, SS_j, SK_j, PK_j)$, and C will then return to the adversary A_1.

4) Extract-partial-private-key-oracle-query: C looks for SS_j in the $l_c - list$ using the input ID_j.
5) Replace-public-key-oracle-query: C looks for the relevant public key PK_j and replaces it with PK'_j when given ID_j and a new public keyPK'_j.Finally, C will subsequently register this replacement.
6) E_{11}, E_{12} oracle-query: On input of $Q_j = (F_h, \vec{\mu_{j-1}}, \vec{\alpha_{j-1}}, sign_{j-1})$,the challanger C arbitrary choose $e_{j,1}, e_{j,2} \in \{0,1\}^\ell$ and them comes back $e_{j,1}, e_{j,2}$ to adversary A_1.
7) G_{F_h} oracle-query: The challanger C selects $g_{j,1}$ or $g_{j,2} \in \{v \in R_p, ||v|| \leq \sigma\sqrt{\frac{N}{2}}\}$ at random from the input of $e_{j,1}\&e_{j,2}$ for $1 \leq j \leq k$, then returns $g_{j,1}$ or $g_{j,2}$ to A_1.
8) H_3-oracle-query: $(y_{j,1} + y_{j,2} * h, \mu_j, c_j)$ are on the $l_{H_3}-list$. C searches for $(y_{j,1}+y_{j,2}*h, \mu_j)$ in the $l_{H_3}-list$ using them as input. C returns the corresponding c_j if they are discovered in it.If not, C picks c_j at random from D_{H_3} and stores $(y_{j,1} + y_{j,2} * h, \mu_j, c_j)$ in the $l_{H_3} - list$.The challanger C then sends c_j back to A_1.
9) CL-Agg-Sign-oracle-query: Upon entry of $(\sum_{j-1}, \mu_j, ID_j)$.The challenger C executes the CL-Agg-Sign algorithm to return a signature after first searching the H_1-Oracle query for the $D_j.\sum_j$ Keep in mind that $x_j = \perp$. To create a legitimate signature in this scenario, C can execute the CL-Agg-Sign Algorithm 1.

- *Forgery:* A_1 produces a legitimate forgery with a non-zero probability, $\sum_k^\star = (F_h, \vec{\alpha_k}, \vec{\mu_k}, sign_k)$. When CL-Agg-Verify($\sum j^\star$), which contains $ID_j^\star, \mu_j^\star, sign_j^\star$,and $c_j^\star$, is executed. Whereas $\mu_j^\star$ has never been asked for CL-Agg-Sign-Oracle, $ID_j^\star$ has never been requested for the Create-sensor-Oracle and Extract-Partial-Private-Key-Oracle queries. The SIS problem on the NTRU lattice can be solved using challenge C as shown below.

  After obtaining the forgery signature $(sign_j^\star, c_j^\star)$ on $(\mu_j^\star, \sum_{j-1}^\star)$, C will use the forking lemma to output a new forgery $(sign_j^\bullet, c_j^\bullet)$ on the same $(\mu_j^\star, \sum_{j-1}^\star)$. Hence, we do this by using an equation

$$H_3\Big[z_{j,1}^\star + z_{j,2}^\star * h\Big] = H_3\Big[z_{j,1}^\bullet + z_{j,2}^\bullet * h\Big]$$

$$\begin{aligned}\Big[z_{j,1}^\star + z_{j,2}^\star * h\Big] - PK_j * c_j^\star = &\Big[z_{j,1}^\bullet + z_{j,2}^\bullet * h\Big] \\ &- PK_j * c_j^\bullet\Big[z_{j,1}^\star - z_{j,1}^\bullet\Big] \\ &+ \Big[z_{j,2}^\star - z_{j,2}^\bullet\Big] * h \\ &- \Big(H_1(ID_j^\bullet) + \delta_j P_j\Big) \\ &*\Big[c_j^\star - c_j^\bullet\Big] = 0\end{aligned}$$

  As a result of

$$H_1\Big(ID_j^\bullet\Big) = S_{j,1} + S_{j,2} * h$$
$$P_j = S'_{j,1} + S'_{j,2} * h$$

  we've arrived at

$$\Big[\Big(z_{j,1}^\star - z_{j,1}^\bullet\Big) - \Big(S_{j,1} + \delta_j S'_{j,1}\Big)\Big(c_j^\star - c_j^\bullet\Big)\Big] + \Big[\Big(z_{j,2}^\star - z_{j,2}^\bullet\Big) - \Big(S_{j,2} + \delta_j S'_{j,2}\Big)\Big(c_j^\star - c_j^\bullet\Big)\Big] * h = 0$$

  also, hold. We may conclude that the distribution of $z_{j,1}^\star, z_{j,2}^\star, z_{j,1}^\bullet$ and $z_{j,2}^\bullet$ is, with a high degree of probability, very near to $D_{z^N,\sigma}$. Consequently, we have $[||z_{j,1}^\star||, ||[z_{j,2}^\star||, ||[z_{j,1}^\bullet||, ||[z_{j,2}^\bullet|| \leq 2\sigma\sqrt{N}$. By unfairness $[(z_{j,1}^\star - z_{j,1}^\bullet) - (S_{j,1} + \delta_j S'_{j,1})(c_j^\star - c_j^\bullet)] \leq ||z_{j,1}^\star|| + ||z_{j,1}^\bullet|| + ||(S_{j,1} + \delta_j S'_{j,1})||||(c_j^\star - c_j^\bullet)|| \leq (2\sigma + 2\eta s)\sqrt{N}$ holds & $[(z_{j,2}^\star - z_{j,2}^\bullet) - (S_{j,2} + \delta_j S'_{j,2})(c_j^\star - c_j^\bullet)] \leq ||z_{j,2}^\star|| + ||z_{j,2}^\bullet|| + ||(S_{j,2} + \delta_j S'_{j,2})||||(c_j^\star - c_j^\bullet)|| \leq (2\sigma + 2\eta s)\sqrt{N}$

  The Gaussian sampling min-entropy property claims that, we are aware that a partial private key exists $D_j = S_{j,1} + S_{j,2}*h$ satisfies $H_1(ID_j^\bullet) = S_{j,1}+S_{j,2}*h$ and three demand postulates then we have $[(z_{j,1}^\star - z_{j,1}^\bullet) - (S_{j,1} + \delta_j S'_{j,1})(c_j^\star - c_j^\bullet)]$, $[(z_{j,2}^\star - z_{j,2}^\bullet) - (S_{j,2} + \delta_j S'_{j,2})(c_j^\star - c_j^\bullet)] \neq 0$ has a probability that is greater than 0.80. The SIS problem on the NTRU lattice is therefore solved by $[(z_{j,1}^\star - z_{j,1}^\bullet) - (S_{j,1}+\delta_j S'_{j,1})(c_j^\star - c_j^\bullet)]$, $[(z_{j,2}^\star - z_{j,2}^\bullet) - (S_{j,2}+\delta_j S'_{j,2})(c_j^\star - c_j^\bullet)]$ where $\beta \geq (2\sigma + 2\eta s)\sqrt{N}$. ■

Theorem 2: The new CLSAS technique is existentially unforgeable against any probabilistic polynomial time (PPT) strong types adversary 2 in the random oracle model if the $(p, 3, (2\sigma + 2\eta s)\sqrt{N})$-SIS on NTRU lattice $\Lambda_{h,p}$ is intractable.

Proof: Theorem 1's proof is similar to this one; please see the full version of it. ■

C. Informal Analysis

In this subsection, we analyze potential cybersecurity threats that may affect the proposed CLSAS scheme and outline how these threats are addressed.

- *Key Replacement Attack:* In certificateless cryptography, adversaries may attempt to replace a user's public key to forge valid signatures. Our scheme resists such attacks through the binding of the public key PK_j to the user identity ID_j via secure hash functions H_1 and H_2, which are modeled as random oracles. This ensures that replacing the public key without knowledge of the partial private key remains computationally infeasible.
- *Replay Attacks:* Replay attacks involve the reuse of previously intercepted messages or signatures. In CLSAS, each signature is generated using fresh ephemeral values $(y_{j,1}, y_{j,2})$ and challenge values c_j derived from a hash of the message and signature components. The use of nonces and random values guarantees freshness, making replay attacks ineffective.
- *Adaptive Chosen Message Attacks (ACMA):* To resist adaptive chosen message attacks, the CLSAS scheme leverages the hardness of the SIS problem on the NTRU lattice. Even if an adversary queries the signing oracle multiple times, without solving the SIS problem, they cannot forge a valid signature on a new message.
- *Partial Key Exposure and Insider Attacks:* An insider with access to the key generation process may attempt to reconstruct full private keys. CLSAS mitigates this

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TABLE I
CRYPTOGRAPHIC OPERATIONS TIME

Operation	Executive Time (μs)
Polynomial Addition T_+	18.7
Polynomial Multiplication T_*	17.9
Polynomial Gaussian T_G	15.67
Hash Function T_h	13.2
Lattice-based encryption T_E	11.32
Lattice-based decryption T_D	11.32

TABLE II
COMPARISON OF COMPUTATIONAL COSTS

Phase	CL-Agg-Sign	CL-Agg-Verify
Xie *et al.* [21]	$k \cdot (T_G + 3T_h + 2T_* + 2T_+)$	$k \cdot (T_G + 2T_h)$
Xie *et al.* [27]	$k \cdot T_G + 6T_h + 1 \cdot T_E + 6T_* + 6T_+$	$k \cdot T_G + T_D + 4T_h$
Xu *et al.* [29]	$k \cdot (T_G + 6T_h + 2T_* + 2T_+)$	$k \cdot (T_G + 1T_h)$
Wang *et al.* [30]	$k \cdot T_G + 3T_h + T_E + 4T_* + 4T_+$	$k \cdot T_G + T_D + 4T_h$
Wang *et al.* [31]	$k \cdot T_G + 4T_h + T_E + 3T_* + 4T_+$	$k \cdot T_G + T_D + 3T_h$
Liu *et al.* [32]	$k \cdot T_G + 4T_h + T_E + 5T_* + 4T_+$	$k \cdot T_G + T_D + 4T_h$
Proposed	$k \cdot T_G + 3T_h + 1 \cdot T_E + 4T_* + 4T_+$	$k \cdot T_G + T_D + 2T_h$

through the separation of key components: the user's partial private key (from the KGC) and secret value (chosen independently by the user). Thus, compromising the KGC alone is insufficient to generate a valid signature.

- **Quantum Attacks:** CLSAS is constructed over NTRU lattices, leveraging hard lattice problems such as SIS and Ideal-SVP. These problems are considered resistant to known quantum algorithms (e.g., Shor's and Grover's), ensuring that the scheme offers quantum-resilient security.
- *Oracle Manipulation Attacks:* CLSAS defines strict rules for oracle queries, ensuring that adversarial queries are logged, bounded, and appropriately handled. Any deviation (e.g., replacing keys or skipping required hash queries) invalidates the forging attempt under the random oracle model assumptions.

VII. PERFORMANCE ANALYSIS

This section compares the performance of the proposed signature protocol to other existing protocols [21], [27], [29], and [30]. Firstly, we demonstrate the time to complete several crucial processes in the proposed protocol. Second, a detailed comparison of the effectiveness of these regimens is provided. The execution time for the cryptographic operations is considered on the premises of [30]. They have completed an experiment on a hardware platform with the Windows 11 operating system and an Intel Core i7-10700 processor with 16GB of RAM. However, it does not provide a hash function implementation. As a result, we evaluated the IPPCP cryptography library's objective hash function execution times. Table I presents the corresponding outcome of execution time for different operations. The polynomial addition operation (T_+) takes approximately 18.7 μs, while polynomial multiplication (T_*) requires 17.9 μs. The polynomial Gaussian sampling (T_G) has a slightly lower execution time of 15.67 μs. Among the listed operations, the hash function (T_h) is relatively efficient, with an execution time of 13.2 μs. Both lattice-based encryption (T_E) and decryption (T_D) are the most efficient, each taking only 11.32 μs. These timings provide insight into the computational costs associated with the fundamental operations, highlighting that lattice-based cryptographic primitives, despite their strong security properties, are computationally efficient and suitable for time-sensitive applications. Table II represents the computational cost corresponding to the algorithms.

It is important to note that a polynomial in our protocol has a degree of 512. Therefore, to store each coefficient of this polynomial, we use 30 bits. The statistical results show

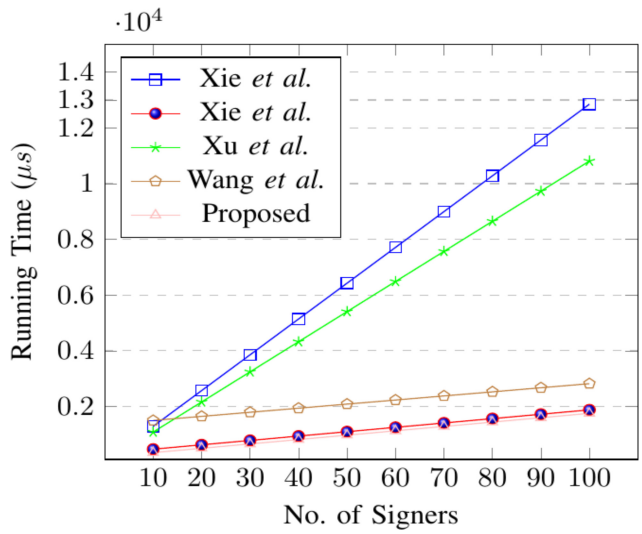

Fig. 2. Time taken by CL-Agg-Sign algorithms for 100 users.

that our Key Generate algorithm contains one hash function, one polynomial Gaussian, one polynomial multiplication, and one polynomial addition. Thus, we can determine that the Key Generate algorithm takes 63.47 μs to complete. In particular, even though our CLSAS has k signers, they can run the signing process concurrently.

The experimental data also show that the estimated number of rejection sampling repeats equals [27]. So, our CL-Agg-Sign algorithm's average running time is $k \cdot T_G + 3T_h + 1 \cdot T_E + 4T_* + 4T_+ = k \cdot 15.67 + 197.32\mu s$. The CL-Agg-verify in our protocol needs $k \cdot T_G + T_D + 2T_h = k \cdot 15.67 + 37.72\mu s$ to confirm the accuracy of the sequential aggregate signature. It is important to note that there is no need to compare the Setup and key generation algorithm's execution time because MS may perform the Setup and key generation algorithms offline.

The polynomial degree is taken 512 for the performance comparison of various existing algorithms with the proposed scheme. Also, we have considered 100 signers for evaluating the performance of algorithms. Therefore, results are computed based on Table I and Table II. Here, Fig. 2 represents the time taken by the CL-Agg-Sign algorithm for 100 signers, and Fig. 3 represents the time taken by the CL-Agg-Verify algorithm for 100 signers. Lucidly, it can be observed from these figures that the total time taken by the proposed mechanism is lesser than the other schemes. Hence, our scheme performs significantly better than other existing protocols.

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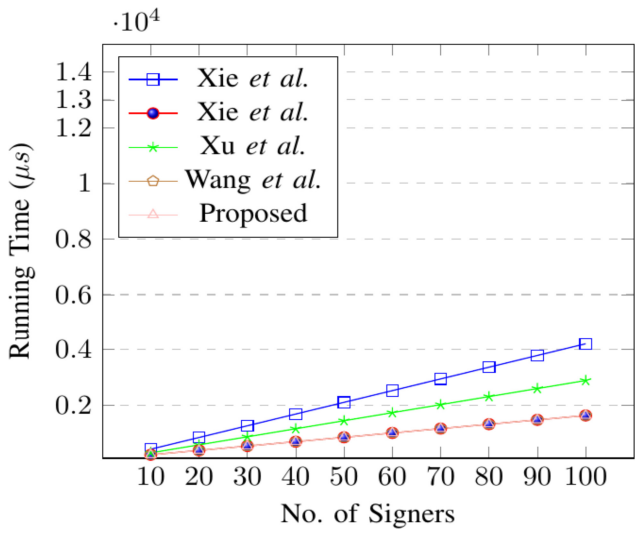

Fig. 3. Time taken by CL-Agg-Verify algorithms for 100 users.

A. Efficiency

Efficiency in the context of sequential aggregate signatures in cryptography typically refers to the performance of generating, verifying, and aggregating signatures in a sequential order. Three certificateless signature (CLS) techniques from the lattice secure random oracle model are presented in references [21], [29] and proposed work. Three certificateless signature (CLS) techniques from the lattice secure random oracle model are presented in references [21], [29] and proposed work.

1) Compression Rate Analysis: The compression rate, in the context of the given signature schemes, refers to the reduction in size achieved by aggregating k signatures into a single aggregated signature compared to the total size of k individual signatures. It measures how effectively the new scheme compresses the signatures. The compression rate is denoted as rate $\{K\}$ and is calculated as follows:

$$rate\{k\} = 1 - \frac{4N\log 2\sigma + Nk(4N\log 2\sigma - \log p) + \eta + 4\ell}{k(4N\log 2\sigma + \eta)}$$

- **Derivation of** $rate\{k\}$**:**
  Expression of $rate\{k\}$ is $rate\{k\}$
  $$= 1 - \frac{4N\log 2\sigma + Nk(4N\log 2\sigma - \log p) + \eta + 4\ell}{k(4N\log 2\sigma + \eta)}$$
  $$\geq 1 - \left(\frac{1}{k} + \frac{4N(\log 2\sigma + \eta - \log p^{\frac{1}{4}}) + 4\ell/k}{4N\log 2\sigma}\right)$$
  $$\geq 1 - \left(\frac{1}{k} + \frac{o(\log p)}{o(\log p + \log(p^{\frac{1}{4}}))}\right)$$
  This step involves substituting the signature size expressions into the formula for $rate\{k\}$ and simplifying.
- *Approximation:* When $p = N^{2c}$ for some c, the term $\log p^{1/4}$ becomes negligible compared to $\log p$. Hence, the expression simplifies to $\text{rate}(k) \approx 1 - \frac{1}{k} - \frac{1}{c}$.

As a result, the compression rate increases until it reaches the asymptotically superior value of $1 - 1/k$, at which point the average size of aggregate and individual signatures will be the same. Secondly, we let $p = poly(N)$. By an inconsequential calculation utilizing $p = N^{2k}$ we have $rate\{k\} \approx 1 - \frac{1}{k} - \frac{1}{c}$, implying that every underwriter produces $1 - 1/k$ of above per signature. Accordingly, we get an $(1 - 1/k)$-CLSAS scheme. Thus, picking k huge enough returns a practically ideal CLSAS scheme. The efficiency comparison among various schemes is shown in Table III.

TABLE III
EFFICIENCY COMPARISON AMONG VARIOUS SCHEMES

Scheme's Name	Signers Number	Signature Size
Xie *et al.* [21]	1	$4N\log 2\sigma + \eta$
Xu *et al.* [29]	1	$3N\log 2\sigma + 2\eta$
Our scheme	k	$2N\log 2\sigma + Nk(\log\sigma - \log p) + \eta + 2\ell$

VIII. CONCLUSION

Considering the impact of a quantum computer on the security of conventional problems and the essential security needs for deploying an CCIoMT network, we have proposed a lattice-based digital signature scheme in this article. A sequential aggregate signature is utilized in the presented protocol, which is helpful for the resource-restrained environment and reduces the computational expanses of the protocol. Also, the proposed protocol concurrently enjoys the attributes of a public key and identity-based cryptosystem because of the employment of a certificateless cryptosystem in the construction of the protocol. Security analysis of protocol demonstrates the security and competence of the protocol under random oracle against active and passive adversaries. The assessment of the existing protocols deduces the projected protocol's computational efficiency in contrast to others. Additionally, the protocol is constructed with the SIS problem of the lattice, which makes it withstand quantum attack. Therefore, the projected protocol has better security and operability, which makes it suitable for the IoMT environment.

In future work, we aim to optimize the proposed scheme for ultra-resource-constrained consumer electronics in IoMT environments. Integration with edge AI will be explored to enable intelligent access control and anomaly detection. We plan to enhance the energy efficiency of lattice operations for battery-powered medical devices. The scheme's scalability to large-scale IoMT networks with frequent signature aggregation will be studied. Support for dynamic device join/leave in real-time scenarios is another area of interest. We also intend to incorporate privacy-preserving techniques such as zero-knowledge proofs. Interoperability with blockchain for secure and verifiable data sharing will be considered. Further efforts will be made to reduce latency and communication overhead. The scheme's resilience to emerging quantum attack models will be rigorously evaluated. Finally, we plan to test the practicality of our approach through real-world deployment and benchmarking on medical IoT platforms.

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